Question 2

A1 Use a Hall probe/search coil/current balance/magnetic field sensor with datalogger or meter to measure magnetic field **1**

A2 Hall probe connected to a galvanometer/microammeter/voltmeter/calibrated Hall probe/datalogger. **1**
Search coil connected to ballistic galvanometer/datalogger.
Current balance and measure force or current (n/a Newton meter)
If magnetic field sensor used then the meter must be specified. Allow datalogger

A3 Insert/remove search coil from field **1**
Arrange plane of Hall probe to be perpendicular to field.
Current balance with some detail (e.g. wire perpendicular to field or adjust current until balance)

B1 Use an appropriate thermometer to measure temperature in the range 0 – 200°C **1**
e.g. thermocouple thermometer/digital thermometer/thermistor thermometer resistance thermometer/electric thermometer
Do not allow vague 'thermometer'.
If mercury-in-glass thermometer employed, then the range must be specified.

B2 Method of heating the magnet uniformly **1**
e.g. oven/constant temperature enclosure/heat in oil
Do not allow the magnet to be heated in boiling water/use of Bunsen flame
Do not allow vague answers such as 'heat source'.
Do not allow magnet to be heated in coil.

B3 Leave magnet for a 'long time' in order to achieve uniform temperature **1**

C1 Vary temperature of magnet and measure B and θ **1**
Only apply to workable arrangements such as Hall probe/search coil/induced e.m.f./ induced current/current balance/force on current-carrying wire.
Do not allow iron filings methods/paper clips/force on a nearby magnet

C2 Keep the distance of the Hall probe to the magnet constant **1**
Allow 'distance from magnet to 'detector' is the same'. Allow 'move magnet at same speed' (if magnet has been inserted into coil).
Unworkable methods cannot score this mark.

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D Any good further design features. Some of these might be: **3**
Perform the experiment away from other magnetic materials
Use datalogger to record/display results (possible three marks for good description)
Awareness that Hall probe must not become hot during the experiment
Allow magnet to reach thermal equilibrium and then quickly place Hall probe next to magnet; record B and then quickly remove the probe.
Attach thermocouple to magnet/thermocouple detail.
Detail relating to calibration of Hall probe.
Use fridge/freezer/ice bath to achieve 0 ^{O}C.
Any good safety point.

Allow other valid points.

11 marking points, but only **10 marks maximum can be scored.**

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